

TOSSAPORN SAENGJA

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EDUCATION

Massachusetts Institute of Technology

Feb 2018 - Present

Masters of Engineering in Computer Science and Engineering

Massachusetts Institute of Technology

Sep 2015 - Feb 2018

Bachelor of Science in Computer Science and Engineering

GPA: 4.8/5.0

AWARDS

4th, MIT ZERO, ACM Boston Preliminary 2016 (BOSPPE) from 37 teams over 16 schools

85th, Bronze, International Olympiad in Informatics 2013 (IOI) from 299 contestants over 77 countries

RESEARCH EXPERIENCE

MIT Media Lab, Human Dynamics

2018–Present

The group uses the tools of computational social science to ask how we can better organize society, government, and companies.

- **Learning Quadratic Games on Large-Scale Network**

Prepared large-scale, real-world datasets for optimization framework to learn the underlying graph that represent network structure and individual marginal benefits

- **Measuring Heterogeneous Social Influence within Homophilous Communities**

Built probabilistic models to quantify heterogeneous influences underlying decision cliques

MIT Media Lab, Big Data – Applied Machine Learning in Andorra, *Researcher* 2016–2018

The project studies the relationship between tourism and energy consumption in Andorra by applying machine learning models on mobile phone record data provided by Andorran Government.

- Performed quantitative analysis on the raw 423GB call detail records (CDRs) and developed a tool that extracts discriminating features from the data using python and pandas.
- Built graphical models performing link prediction and analysis of friending between two strangers.
- Built various regression models on top of the features extracted from the CDR datasets with scikit-learn and statsmodels.

WORK EXPERIENCE

DeWolfe Group, Technical Cofounder

2018

DeWolfe tries to drive mainstream adoption of blockchain in real life with the main project being a universal payments platform which allows users holding any cryptocurrency to make direct purchases at real-world merchants.

- Designed and built mobile applications for alpha testing using Sketch and React Native.
- Developed the backend system to work with multiple cryptocurrency exchange platforms using Type-script, Postgres, and RabbitMQ.
- Developed the backend system to communicate with Bitcoin and Ethereum protocols.

Facebook, Notification Backend, *Intern*

2017

The team mainly use machine learning models to get notifications scores for further use such as dropping noisy notifications or ranking them by importance.

- Worked with Hive tables of terabytes size.

- Implemented a production pipeline for network effect prediction models, both classification and regression, using HiveQL and FBLearner Flow within Facebook's machine learning framework.
- Added support and introduced regression model in notification backend framework.
- Increased 0.5% comments metrics in production with 2 billions active users at the time.

PROJECTS

A Song for Me: A Music Sharing Tool and Its Evaluation 2019

MIT MAS.S60: Artificial Intelligence and Mental Health

Built a Chrome extension and conducted a study of it. The extension lets the users create the "Up Next" feature on YouTube with handpicked songs from their friends. Preliminary survey results from 13 participants show positive effects on the relationships of the users. The project uses scikit-learn, Django, Nginx, React, and Javascript.

Network Wrangler 2019

MIT 6.894: Interactive Visualization

Built an interactive network data processing and visualization tool. Features include node selections, filtering using brushing techniques, and multi-view layouts. It is developed using React, React-vis, and VivaGraphJS.

OmiseGO Token Sales Study 2018

MIT MAS.S62: Cryptocurrency Engineering and Design Scraped the transactions on the Ethereum network and performed data analysis to characterize, understand, and identify patterns centered around the token sales.

Accurate Clinical Decision Support Systems 2017

MIT 6.S897: Machine Learning for Healthcare

Used logistic regression, random forest, and Latent Dirichlet allocation to analyze the reasons behind dismissals of clinical decision support systems on the real dataset from Brigham and Women's hospital.

iMake 2017

MIT 6.813: User Interface Design & Implementation

Designed and implemented a shopping website that utilizes cube-like view and "hotspots" for the items using React.

Paraphrase Identification 2016

MIT 6.867: Machine Learning

Designed and implemented learning models for paraphrase identification as a binary classification task. The models' architectures include bag-of-words, word-embeddings, support vector machines, neural networks with dynamic pooling layers, and long short-term memory networks. Acquired the best accuracy, F1 score of 71.7% and 81.8% respectively on Microsoft Research Paraphrase Corpus.

TEACHING

Teaching Assistant for MIT 6.824: Distributed Systems, Spring 2020

Lab Assistant for MIT 6.s080: Brief Intro to Python, IAP 2017

Student Lab Assistant for MIT 6.01: Intro to Electrical Engineering and Computer Science, Fall 2016

Lecturer for Thailand Regional Computer Olympiad Camp, 2013-2014

LEADERSHIP

President for Thai Students at MIT in 2017

Treasurer for Thai Students at MIT in 2016